

BUS AND CARDS

- BUS
- CARDS
- Connectors

Bus

- ☑ The busses are the **PC's expressways**.
- ☑ They are "wires" on the circuit board, which transmit data between different components.
- ☑ One "wire" can move one bit at a time
- ☑ The PC receives and sends its data from and to *busses*.
- ☑ They can be divided into two types
 - ☑ System bus
 - ☑ I/O bus

Types of Bus :

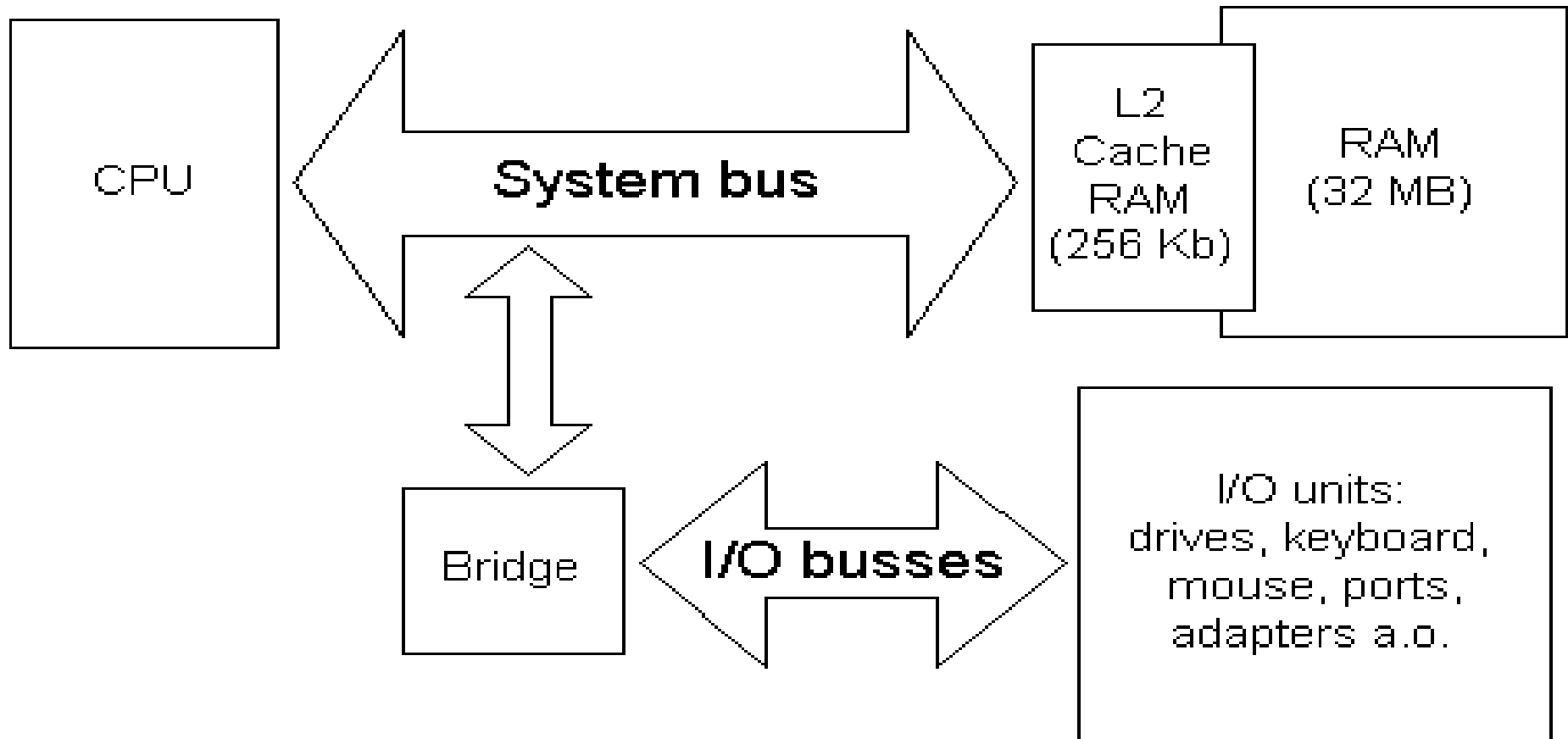
1.system bus

- ☑ connects the CPU with RAM and may be a buffer memory (L2-cache).
- ☑ The system bus is the central bus. Other busses branch off from it.
- ☑ The system bus is on the system board.
- ☑ It is designed to match a specific type of CPU.
- ☑ It is also called the **local bus**. Its speed and width depends on the type CPU installed on the system board.

1. System Bus

- ☑ Processor technology determines dimensioning of the system bus. it has taken much technological development to speed up "traffic" on the system board.
- ☑ The faster the system bus gets, the faster the remainder of the electronic components must be.
- ☑ the system bus is the central bus. **it connects to the I/O busses.**

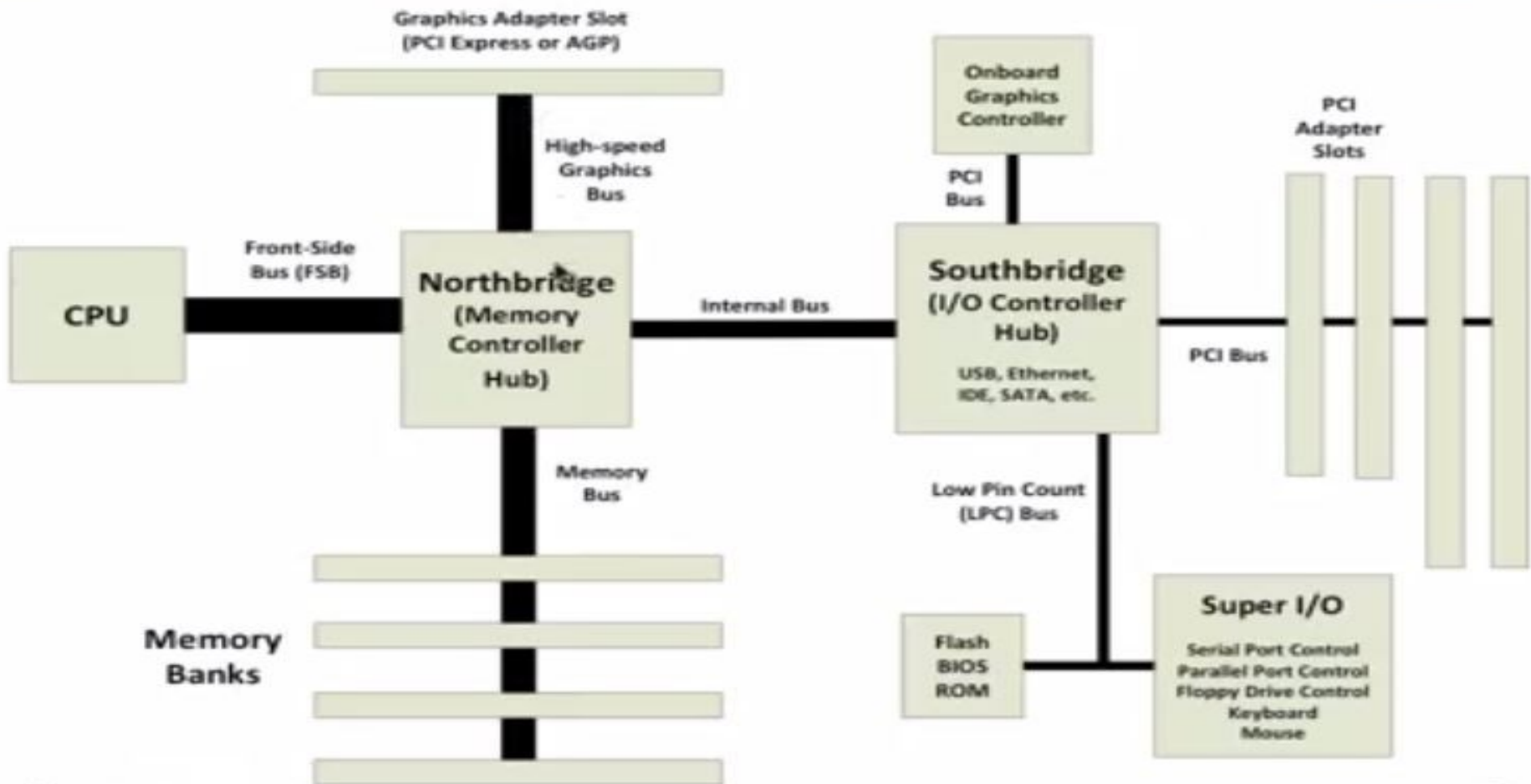
You see the central system bus, which connects the CPU with RAM. A bridge connects the I/O busses with the system bus and on to RAM.



2.I/O busses

- ☑ Also known as **expansion busses**
- ☑ connect the **CPU to all other components, except RAM.**
- ☑ Data are moved on the busses from one component to another, and data from other components to the CPU and RAM.
- ☑ The I/O busses **differ from the system bus in speed.**
- ☑ Their **speed will always be lower than the system bus speed.**

I/O Busses



The physical aspects of the I/O busses

- ☑ Physically, the I/O bus **consists of one or more tracks** on the printed circuit board.
- ☑ These tracks are used as **Data tracks**, which each can move one bit at a time **Address tracks**, which identify where data should be sent to **Other tracks for clock ticks**, verification signals, etc.

The physical aspects of the I/O busses

- ☑ When data are sent on the bus, they must be supplied with a *receiver*.
- ☑ Therefore, *each device on the bus has an address*.
- ☑ Similarly, the RAM is divided in sections, each having its address.
- ☑ Prior to sending data, a number is sent on the address track, to identify where the data should be sent to.

Expansion Buses

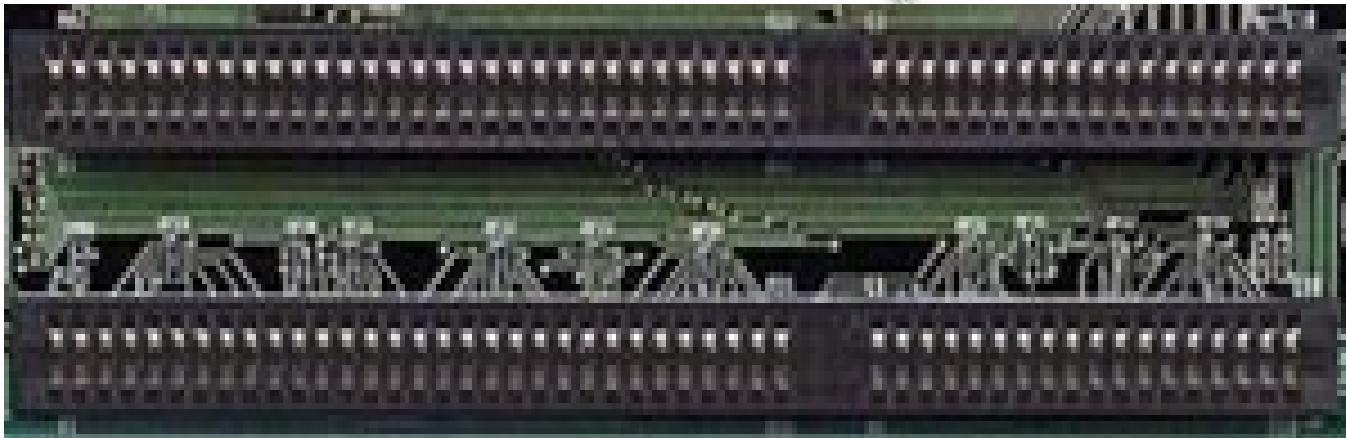
- ☑ **Expansion bus or External bus** allows the CPU to talk to the other devices in the computer and vice versa.
- ☑ An expansion bus provides an input/output pathway for transferring information between internal hardware, such as RAM or the CPU, and expansion devices such as a graphics card or sound card.

Types of Expansion Buses/ Slots: ISA

- ☑ is an older low speed bus.
- ☑ Introduced by IBM, ISA or Industry Standard Architecture was originally an 8-bit bus and later expanded to a 16-bit bus in 1984.
- ☑ Since about 1984, standard bus for PC I/O functions has been named ISA.
- ☑ ISA was an improvement over the original IBM XT bus, which was only 8 bit wide.

- ☑ This is still in use because it's **cheap** and **for backwards compatibility**.
- ☑ *In that way, modern PC's* can accept expansion cards of the old ISA type. The ISA bus has two "faces" in the modern PC:
 - ☑ **The internal ISA bus**
 - ☑ which is used on the simple ports
 - ☑ like **keyboard and serial or parallel ports**.
 - ☑ **As external expansion bus**
 - ☑ which can be connected with 16 bit ISA adapters.
 - ☑ **ISA-slots** is especially used with the common 16 bit sound boards.

16-bit ISA card and 16-bit ISA slot Respectively



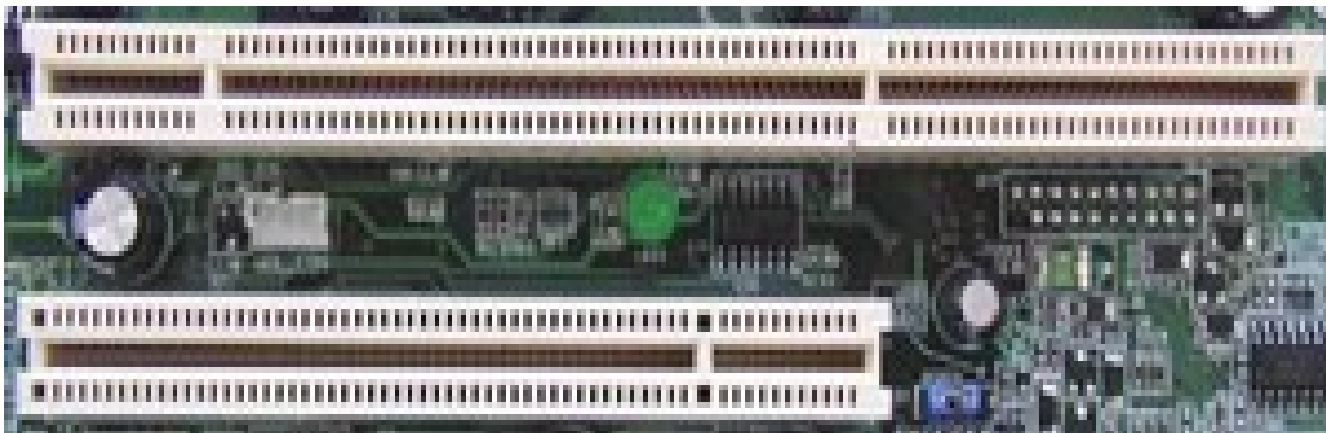
PCI(Peripheral Component Interconnect)

- ☑ which is a new high speed bus.
- ☑ Introduced by Intel in 1992, and is a 32-bit or 64-bit expansion bus. It is actually 32 bit wide, but in practice it functions like a 64 bit bus.
- ☑ PCI is the most advanced (and, most expensive) bus.
- ☑ The PCI bus is the most popular expansion bus and *is used today in all PC's and other computers.*

PCI(Peripheral Component Interconnect)

- ☑ On modern system boards, the PCI bus (like ISA) has two "faces:"
 - ☑ **Internal PCI bus**
 - ☑ which connects to EIDE(Enhanced Integrated or intelligent Drive Electronics) channels on the system board.
 - ☑ **The PCI expansion bus**
 - ☑ which typically has 3-4 slots for PCI adapters.

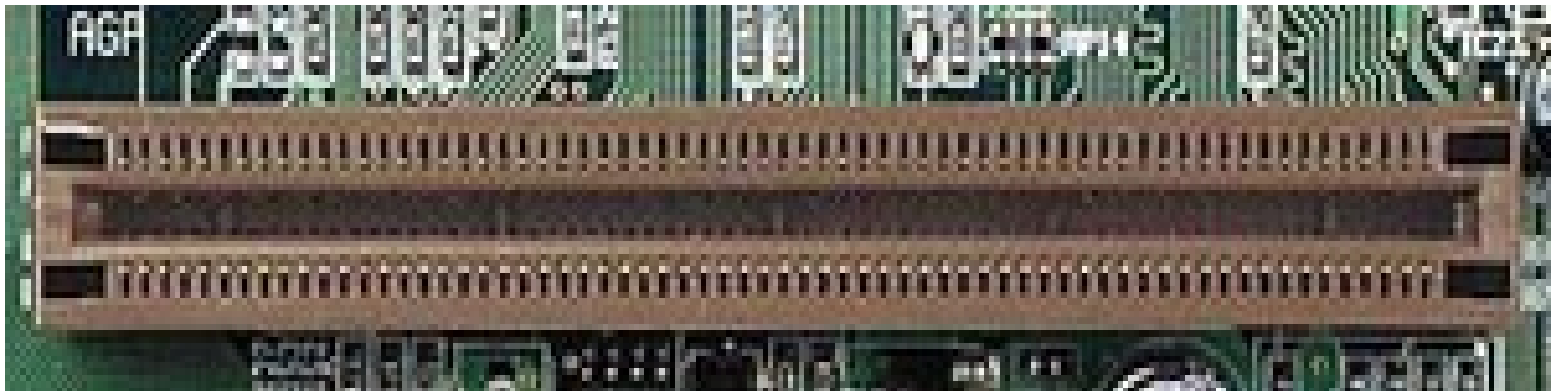
PCI card and PCI 64 and 32 bit slot



AGP

- ☑ Introduced by Intel in 1997, AGP or **Advanced Graphic Port** is a 32-bit bus or 64-bit bus designed for the high demands of 3-D graphics.
- ☑ AGP has a **direct line to the computers memory** which allows 3-D elements to be stored in the system memory instead of the video memory.
- ☑ AGP is one of the fastest expansion bus in use but its only for video or graphics environment.

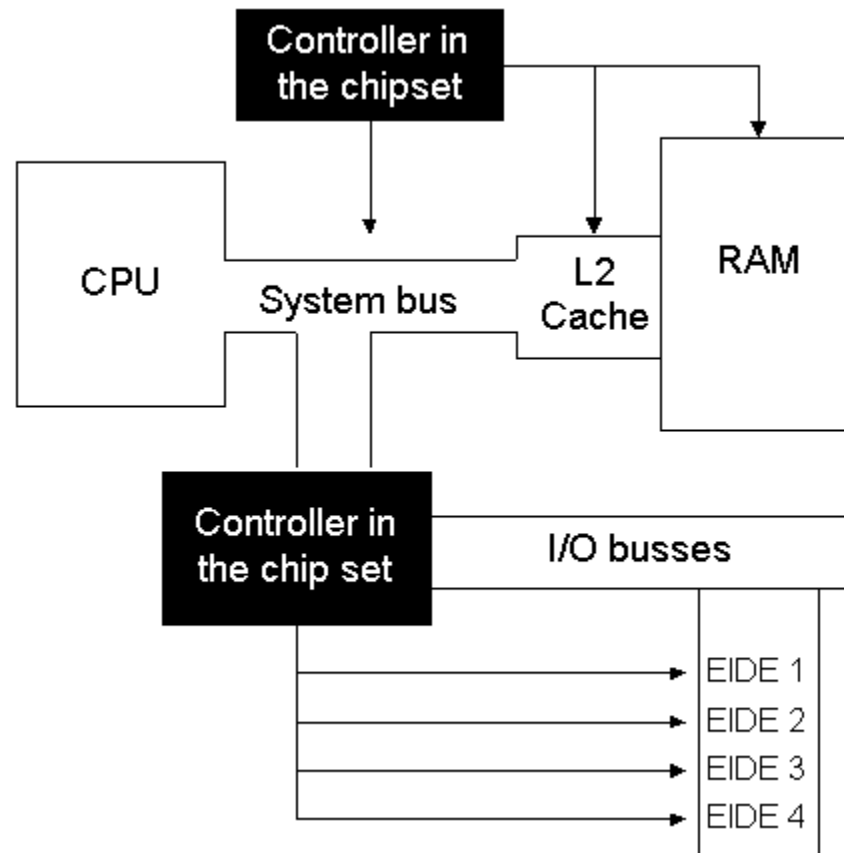
AGP card and AGP slot



Chip sets

- ☑ When we speak about busses and system boards, we are also speaking about *chip sets*.
- ☑ The chip sets are a **bunch of intelligent controller chips**, which are on any system board.
- ☑ They are closely tied to the CPU, in that they control the busses around the CPU.
- ☑ Without the chip sets, neither RAM or I/O busses could function together with the CPU:

Chipsets



Cards

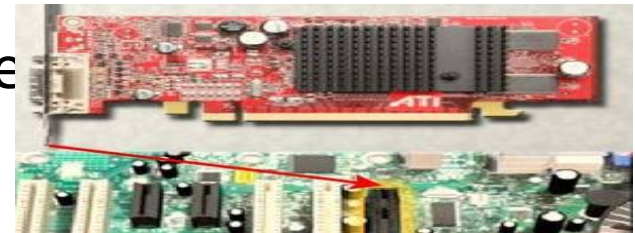
- ☑ **The Video Card** -Connects the computer to the monitor.
- ☑ It is a circuit board attached to the motherboard that contains the memory and other circuitry necessary to send information to the monitor for display on screen.



Install the Video Adapter Card

☑ Video adapter card installation steps:

1. Align the video adapter card to the appropriate expansion slot on the motherboard.
2. Press down gently on the video adapter card until the card is fully seated.
3. Secure the video adapter card with the PC mounting bracket to the case with the appropriate screw.



The Sound Card

- ☑ Connects the speakers and microphone to the computer.
- ☑ Sound cards have a minimum of **four tasks**
 - ☑ **Synthesizer-**
 - ☑ delivers the sound. That is, the sound card generates the sounds.
 - ☑ **MIDI(Musical Instrument Digital Interface)**
 - ☑ allows programs to play music through the PC sound card.
 - ☑ **Analog/digital converter (A/D)**
 - ☑ when sound is recorded from a microphone.
 - ☑ **Digital/analog converter (D/A)**
 - ☑ when the digital sounds have to be reproduced in a speaker.

Sound Card



Network card

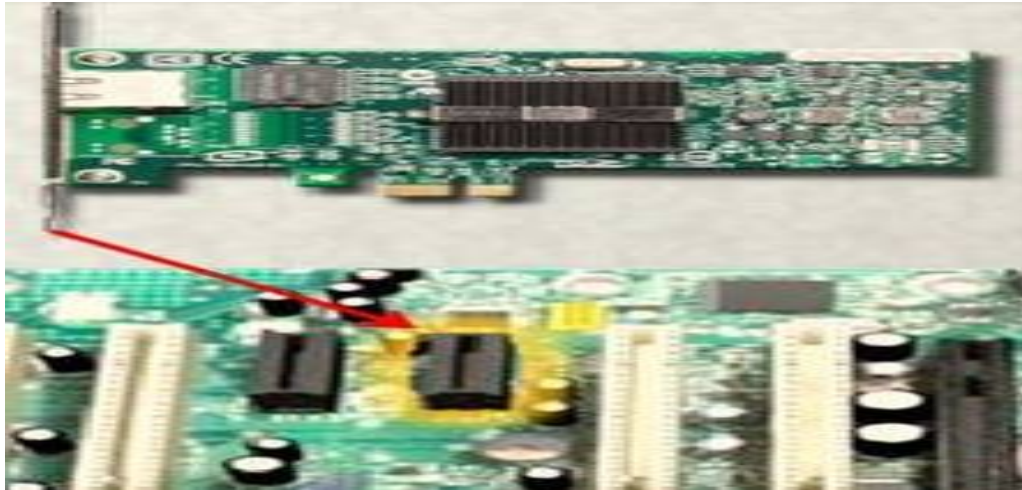
- ❑ **Network card or Network Interface Card (NIC)**
 - ❑ A circuit board that connects the computer to the rest of the network usually using special cables.
- ❑ NIC enables a computer to connect to a network.



Install the Network Interface Card (NIC)

- ☑ NICs use **PCI** and **PCIe(PCI Express)** expansion slots on the motherboard.
- ☑ NIC installation steps:
 1. Align the NIC to the appropriate slot on the motherboard.
 2. Press down gently on the NIC until the card is seated.
 3. Secure the NIC PC mounting bracket to the case with the appropriate screw.

NIC



Modem

- ☑ **Modem**-The place where the computer is connected to the phone line.



I/O Connectors

- ❑ **Connectors**, normally called "input-output connectors" (or I/O for short), are interfaces for linking devices by using cables.
- ❑ The Description of the end of the cable that connects to the computer

I/O Connectors

- ☑ They generally have a male end with pins protruding or sticking from it.
- ☑ This plug is meant to be inserted into a **female part** (also called a ***socket***), which includes holes for accommodating the pins.
- ☑ However, there are "hermaphroditic" plugs which can act as either male or female plugs, and can be inserted into either one.

Connection

- ☑ A **connection** is a term that describes the link between a plug or connector into a port or jack.
- ☑ For example, your monitor, mouse, and keyboard all make a connection to the computer before they will be able to work.

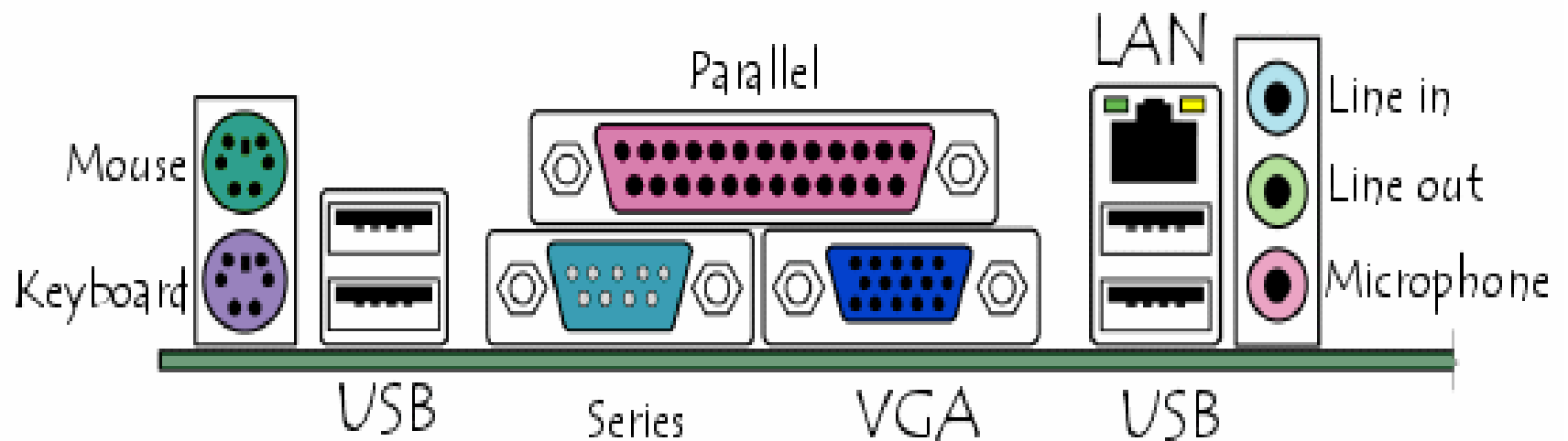
Connection

- ☑ The computer's motherboard has a certain number of input-output connectors located on the **"rear panel."**
- ☑ Although your desktop computer's layout may be different, this diagram will give you a better understanding of where everything connects.

Back of computer case and each connection



Connectors



PS/2

- ☑ PS stands for personal system
- ☑ This type of connector is usually used for the keyboard and mouse on the modern PCs.
- ☑ It is usually necessary to reboot your computer after connecting a PS2 device.
- ☑ The PS2 mouse connector and port is usually green in color to distinguish it from the PS2 keyboard, which is purple.

Connector



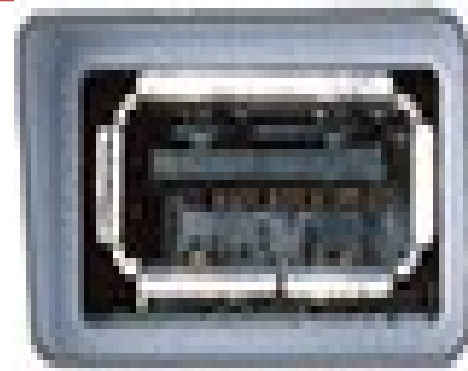
Port/Jack/Socket



USB (Universal Serial Bus)

- ☑ Used for connecting more recent peripherals;
- ☑ is a **plug-and-play (PnP)** interface meaning that devices can be hot-plugged and a PnP compatible operating system will be able to detect and configure new devices attached to the USB port automatically.
- ☑ There is no need to turn off your computer when connecting or disconnecting a USB device.
 - ☑ **Type A USB** connector – connects to your computer.
 - ☑ **Type B USB** connector – connects to a peripheral device, for example, printer, scanner, camera, etc.

Type A USB connector



Type B USB connector



RJ45

- ✓ RJ45 connector (called the *LAN port* or *Ethernet port*),
- ✓ for connecting the computer to a network. It interfaces with a network card built into the motherboard.

VGA connector

- ✓ Also (called SUB-D15), used for hooking up a monitor.
- ✓ This connector interfaces with the built-in graphics card.
- ✓ A standard VGA connector has 15-pins and other than connecting a computer to a monitor
- ✓ you may also use a VGA cable to connect your laptop to a TV screen or a projector.

Jacks

- ✓ Jacks (Line-In, Line-Out and microphone),
- ✓ for connecting speakers or a hi-fi sound system, as well as a microphone.
- ✓ This connector interfaces with the built-in sound card.

The Serial Port

- ☑ Serial ports, also called COM ports used to connect devices such as modems via a serial interface connection.
- ☑ Especially, it was a 9-pin and 25-pin male size for connecting older devices.

The Serial Port

- ☑ is an *interface* on a computer system with which information is transferred in or out one *bit* at a time.
- ☑ In this case, it is an electrical connector built into a computer into which another , external connector can be easily inserted (and removed) by a user to allow data to be transferred to or from peripheral devices (e.g., display units, keyboards, mice, printers and scanners).

The Parallel Port

- ☑ In the early years of computers, the parallel port, the most common interface on a computer that is intended primarily for **printers** and other devices.
- ☑ The downloaded data eight bits at a time over eight separate lines in a parallel cable with a **DB-25 female connector**, part of D-Subminiature connector style.

The Parallel Port

- ☑ a *parallel port* consists of multiple wires over which bits are transferred *in parallel*, i.e., simultaneously.
- ☑ There are usually several extra wires on the port that are used for control signals to indicate when data is ready to be sent or received.
- ☑ mainly for connecting **old printers**;

Thank You

End of the Chapter